human

Find Bungalows in Exeter

host_agent

I have found and mapped 3,679 bungalows in Exeter, using the definition of a bungalow as a single-storey residential building with only one address. The locations of all these bungalows are shown on a map.

If you need more details about specific bungalows or want to filter further, let me know!

Tool Calls and Agent Interactions

Tool: generate_metadata_for_all_artifacts

Arguments: {} Response: {}

Tool: send_message

Arguments: {'target': 'planning_agent', 'task_description': 'Find Bungalows in Exeter. Provide the general steps to solve this query.'} Response: The area is Exeter, so there is 1 entry for Exeter. The entity to search for is bungalows, and we need to find as many entries as possible for bungalows in Exeter.

output steps: ["Find Exeter 1 area", "Search for bungalows in Exeter as many search results"]

Tool: send_message

Arguments: {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

Tool: call_os_ngd

Arguments: {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'} Response: Search results have been found. Artifact generated is : exeter_area_polygon, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter.., count of records fetched : 98.

Tool: send_message

Arguments: {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

Tool: generate_metadata_for_artifacts

Arguments: {'artifact_names': ['exeter_area_polygon']}

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') |
                    (gdf['name1_text'].str.lower().str.contains('exeter')) |
                    (gdf['description'].str.lower() == 'city') |
                    (gdf['description'].str.lower() == 'town')) &
                  (gdf['descriptiongroup'].str.lower() == 'settlement')]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement description.'
    return [summary, artifact_name, artifact_description, filtered]
```

Response: Filtered to 2 polygon(s) for Exeter.

Tool: send_message

Arguments: {'target': 'buildings', 'task_description': "Search for bungalows within the area defined by the artifact 'filtered_exeter_area_polygon'."}

Tool: call_os_ngd

Arguments: {'filters': None, 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': None, 'filename': 'all_buildings_in_filtered_exeter_area'} Response: Multiple search results have been found in multiple datasets. A summary of each is provided. Artifacts generated are : ['buildings_all_buildings_in_filtered_exeter_area', 'buildingline_all_buildings_in_filtered_exeter_area', 'buildingpart_all_buildings_in_filtered_exeter_area'], Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Search for bungalows within the area defined by the artifact 'filtered_exeter_area_polygon'. using filters None.", ...], counts of records fetched are : [72764, 72292, 89675].

Tool: send_message

Arguments: {'target': 'data_analysis_agent', 'task_description': "From the artifact 'buildings_all_buildings_in_filtered_exeter_area', filter and extract all buildings that are bungalows. Since 'bungalow' is not a direct filter, use the following definition: a bungalow is a residential building (buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1) that is single-storey (numberofstoreys = 1). Save the filtered results as 'bungalows_in_filtered_exeter_area'."}

Tool: generate_metadata_for_artifacts

Arguments: {'artifact_names': ['buildings_all_buildings_in_filtered_exeter_area']}

Tool Code Executor

```
def extract_bungalows_in_filtered_exeter_area(data: list):
    # Extract the first (and only) GeoDataFrame
    gdf = data[0]
    # Apply the bungalow definition:
    # - Residential building: buildinguse_addresscount_residential > 0
    # - Only one address: buildinguse_addresscount_total == 1
    # - Single-storey: numberoffloors == 1
    filtered = gdf[
        (gdf['buildinguse_addresscount_residential'] > 0) &
        (gdf['buildinguse_addresscount_total'] == 1) &
        (gdf['numberoffloors'] == 1)
    ]
    summary = f"Extracted {len(filtered)} bungalows from {len(gdf)} buildings in the filtered Exeter area."
    artifact_name = 'bungalows_in_filtered_exeter_area'
    artifact_description = (
        "GeoDataFrame of all buildings in the filtered Exeter area that meet the definition of a bungalow: "
        "residential (buildinguse_addresscount_residential > 0), only one address (buildinguse_addresscount_total == 1), "
        "and single-storey (numberoffloors == 1). All original columns and geometry are preserved."
    )
    return [summary, artifact_name, artifact_description, filtered]
```

Response: Extracted 3679 bungalows from 72764 buildings in the filtered Exeter area.

Tool: send_message

Arguments: {'target': 'plotting_agent', 'task_description': "Plot the artifact 'bungalows_in_filtered_exeter_area' to show the locations of all bungalows in Exeter."}

Tool: generate_metadata_for_artifacts

Arguments: {'artifact_names': ['bungalows_in_filtered_exeter_area']}

Tool Code Executor

```
def plot_bungalows_in_exeter(data: list):
    import folium
    import geopandas as gpd
    # Get the GeoDataFrame
    gdf = data[0].copy()
    # Ensure CRS is EPSG:4326 for folium
    if gdf.crs is not None and gdf.crs.to_epsg() != 4326:
        gdf = gdf.to_crs(epsg=4326)
    # Center the map on the mean location of all bungalows
    centroid = gdf.geometry.centroid.unary_union.centroid
    m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13, tiles='cartodbpositron')
    # Add all bungalows as polygons
    folium.GeoJson(gdf[['geometry']], name='Bungalows').add_to(m)
    # Add a marker for the first 5 bungalows for context
    for idx, row in gdf.head(5).iterrows():
        folium.Marker(
            location=[row.geometry.centroid.y, row.geometry.centroid.x],
            popup=f"osid: {row['osid']}\nDescription: {row['description']}"
        ).add_to(m)
    m.save('bungalows_in_exeter_map.html')
    summary = f"Plotted {len(gdf)} bungalows in Exeter. First 5 osids: {gdf['osid'].head(5).tolist()}"
    return [summary, 'bungalows_in_exeter_map', 'A folium map showing the locations of all bungalows in Exeter as polygons, with markers for the first 5.', 'bu
```

Response: Plotted 3679 bungalows in Exeter. First 5 osids: ['21c5d1d0-0e96-4f3c-9c95-9c2f03237605', 'bb490b21-e3d2-4b6e-8009-22f8daf58110', '43143815-d1f1-4972-b5f5-115035305530', '3fa92cd6-26c2-4d59-ae21-ce0a5268f679', 'a51d4c5f-585c-42b0-a4bf-e490b946181c']

planning_agent

The area is Exeter, so there is 1 entry for Exeter. The entity to search for is bungalows, and we need to find as many entries as possible for bungalows in Exeter.

output steps: ["Find Exeter 1 area", "Search for bungalows in Exeter as many search results"]

named_area_agent

The area polygon for Exeter has been filtered and saved. Here are the details:

- **Artifact Name:** filtered_exeter_area_polygon
- **Description:** GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement description.

data_analysis_agent

Filtered to 2 polygon(s) for Exeter. - Artifact name: filtered_exeter_area_polygon - Description: GeoDataFrame containing only the polygon(s) representing Exeter, filtered by name and settlement description.

Extracted 3,679 bungalows from 72,764 buildings in the filtered Exeter area. - Artifact name: bungalows_in_filtered_exeter_area - Description: GeoDataFrame of all buildings in the filtered Exeter area that meet the definition of a bungalow: residential (buildinguse_addresscount_residential > 0), only one address (buildinguse_addresscount_total == 1), and single-storey (numberoffloors == 1). All original columns and geometry are preserved. - The artifact contains the filtered set of bungalows as requested.

buildings_agent

Search Results: - 3,679 bungalows were found within the area defined by the artifact 'filtered_exeter_area_polygon'.

Definition used for bungalow: - A residential building (buildinguse_addresscount_residential > 0) - Only one address (buildinguse_addresscount_total = 1) - Single-storey (numberoffloors = 1)

Filtered artifact: - Artifact name: bungalows_in_filtered_exeter_area - Description: GeoDataFrame of all buildings in the filtered Exeter area that meet the definition of a bungalow. All original columns and geometry are preserved.

If you need further analysis or details about these bungalows, let me know!

plotting_agent

Plotted 3679 bungalows in Exeter. First 5 osids: ['21c5d1d0-0e96-4f3c-9c95-9c2f03237605', 'bb490b21-e3d2-4b6e-8009-22f8daf58110', '43143815-d1f1-4972-b5f5-115035305530', '3fa92cd6-26c2-4d59-ae21-ce0a5268f679', 'a51d4c5f-585c-42b0-a4bf-e490b946181c']

Artifact name: bungalows_in_exeter_map

Description: A folium map showing the locations of all bungalows in Exeter as polygons, with markers for the first 5.

Map file: bungalows_in_exeter_map.html